

OBAN, United Kingdom

WMO: 031140

Lat: 56.417N

Lon: 5.467W

Elev: 4

StdP: 101.28

Time Zone: 0.00 (EUW)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.2	-0.7	-6.0	2.3	1.6	-4.7	2.5	1.6	15.5	5.1	13.5	7.3	0.2	160	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	23.4	16.7	21.3	15.7	19.6	15.0	17.7	21.7	16.6	20.0	15.7	18.5	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.4	19.7	15.1	10.7	18.0	14.3	10.2	17.1	49.8	22.0	46.6	20.1	43.8	18.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	10.2	8.7	DB	-4.6	25.9	2.1	1.7	-6.1	27.1	-7.3	28.1	-8.5	29.1	-10.1	30.3	(4)
(5)			WB	-5.1	20.1	2.0	3.0	-6.5	22.3	-7.6	24.0	-8.8	25.7	-10.2	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	5.4	5.5	6.6	8.0	11.6	12.9	15.1	15.0	12.7	9.9	7.5	5.0	(6)	
	DBStd	4.41	2.45	2.55	2.25	2.30	2.90	2.15	2.34	2.46	2.27	2.62	2.83	3.13	(7)	
	HDD10.0	735	143	128	107	67	15	1	0	0	3	34	82	154	(8)	
	HDD18.3	3191	401	361	365	309	210	163	106	109	170	260	325	412	(9)	
	CDD10.0	600	1	1	1	7	64	89	157	156	84	32	7	1	(10)	
	CDD18.3	13	0	0	0	0	1	1	4	7	0	0	0	0	(11)	
	CDH23.3	53	0	0	0	0	2	7	25	19	0	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.8	3.7	3.7	3.3	2.8	2.3	2.3	2.1	2.0	2.2	2.9	3.3	3.1	(14)	
(15) Precipitation	PrecAvg	2151	273	196	182	127	116	116	127	145	172	227	228	263	(15)	
	PrecMax	2846	472	406	453	219	300	221	202	297	334	380	359	473	(16)	
	PrecMin	1448	101	71	31	43	34	41	57	40	44	50	98	55	(17)	
	PrecStd	349	90	103	95	41	63	46	36	62	77	99	75	121	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.1	12.4	16.8	23.4	24.0	25.8	25.4	21.0	17.3	14.3	12.5	(19)	
		MCWB	10.0	10.9	9.1	11.6	14.7	15.5	18.4	18.0	16.9	13.5	12.6	11.2	(20)	
	2%	DB	10.9	11.0	11.0	14.1	20.9	20.6	23.3	23.1	18.6	15.5	13.1	11.4	(21)	
		MCWB	9.8	9.7	9.0	9.8	13.1	14.6	17.3	17.3	14.9	13.0	11.4	10.2	(22)	
	5%	DB	9.9	10.1	10.3	12.6	18.6	18.4	20.8	20.9	17.3	14.6	12.4	10.3	(23)	
		MCWB	8.6	9.0	8.6	9.3	12.5	13.7	15.6	16.7	14.0	12.5	10.9	9.1	(24)	
	10%	DB	8.9	9.2	9.7	11.5	16.4	16.7	19.0	19.0	16.3	13.7	11.5	9.4	(25)	
		MCWB	7.7	8.1	8.2	8.7	11.5	12.6	14.8	15.3	13.5	11.8	10.1	8.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	11.4	10.3	12.2	15.3	17.2	19.6	19.0	17.5	14.2	13.1	11.6	(27)	
		MCDB	11.7	11.8	11.2	15.1	21.7	22.6	23.7	23.8	20.0	15.9	14.0	12.3	(28)	
	2%	WB	9.8	9.9	9.6	10.9	14.0	15.4	17.6	18.0	15.8	13.6	12.0	10.5	(29)	
		MCDB	10.7	10.6	10.4	13.3	19.7	19.1	21.9	22.0	17.8	15.0	12.8	11.3	(30)	
	5%	WB	8.7	9.1	9.1	10.0	12.9	14.3	16.3	16.9	14.7	12.9	11.1	9.3	(31)	
		MCDB	9.6	9.9	9.8	11.8	17.3	17.4	19.8	20.3	16.5	14.3	12.1	10.1	(32)	
	10%	WB	7.9	8.3	8.6	9.2	12.0	13.3	15.3	15.9	13.9	12.1	10.1	8.3	(33)	
		MCDB	8.8	9.1	9.5	10.8	15.7	16.1	18.4	18.5	15.8	13.6	11.4	9.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	3.9	4.2	5.2	7.0	5.9	5.6	5.8	5.9	4.6	4.1	4.1	(35)	
		MCDBR	4.6	3.6	4.7	7.1	10.9	9.2	9.6	8.4	7.1	5.0	4.1	4.5	(36)	
	5% WB	MCWBR	4.6	3.6	3.6	4.5	5.7	5.0	5.2	4.4	4.4	4.0	3.9	4.7	(37)	
		MCWBR	4.4	3.4	3.8	6.0	9.6	8.0	8.1	7.6	6.0	4.3	3.9	4.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.348	0.368	0.382	0.387	0.379	0.383	0.366	0.362	0.346	0.328	0.323	(40)	
		taud	2.179	2.280	2.290	2.301	2.336	2.394	2.392	2.449	2.456	2.412	2.270	2.152	(41)	
	Edn at Noon		521	698	785	833	852	863	851	844	792	701	540	431	(42)	
			59	78	98	112	115	110	108	96	83	68	53	47	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.42	1.05	2.09	3.48	4.55	4.42	4.22	3.35	2.31	1.23	0.55	0.32	(44)	
	RadStd		0.07	0.15	0.18	0.28	0.43	0.50	0.43	0.30	0.19	0.17	0.05	0.04	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page